Questions for further study

1. Visit your local map library, and determine: (1) the projections and datums used by selected maps; (2) the coordinates of your house in several common georeferencing systems.
2. Summarize the arguments for and against a single global figure of the Earth, such as WGS84.
3. How would you go about identifying the projection used by a common map source, such as the weather maps shown by a TV station or in a newspaper?
4. Chapter 14 discusses various forms of measurement in GIS. Review each of those methods, and the issues involved in performing analysis on databases that use different map projections. Identify the map projections that would be best for measurement of (1) area, (2) length, (3) shape.

Further reading

Bugayevskiy L.M. and Snyder J.P. 1995 *Map Projections: A Reference Manual*. London: Taylor and Francis.

Kennedy M. 1996 *The Global Positioning System and GIS: An Introduction*. Chelsea, Michigan: Ann Arbor Press.

Maling D.H. 1992 *Coordinate Systems and Map Projections* (2nd edn). Oxford: Pergamon.

Sobel D. 1995 *Longitude: The True Story of a Lone Genius Who Solved the Greatest Scientific Problem of His Time*. New York: Walker.

Snyder J.P. 1997 *Flattening the Earth: Two Thousand Years of Map Projections*. Chicago: University of Chicago Press.

Steede-Terry K. 2000 *Integrating GIS and the Global Positioning System*. Redlands, CA: ESRI Press.

6 Uncertainty

Uncertainty in geographic representation arises because, of necessity, almost all representations of the world are incomplete. As a result, data in a GIS can be subject to measurement error, out of date, excessively generalized, or just plain wrong. This chapter identifies many of the sources of geographic uncertainty and the ways in which they operate in GIS-based representations. Uncertainty arises from the way that GIS users conceive of the world, how they measure and represent it, and how they analyze their representations of it. This chapter investigates a number of conceptual issues in the creation and management of uncertainty, before reviewing the ways in which it may be measured using statistical and other methods. The propagation of uncertainty through geographical analysis is then considered. Uncertainty is an inevitable characteristic of GIS usage, and one that users must learn to live with. In these circumstances, it becomes clear that all decisions based on GIS are also subject to uncertainty.

Geographic Information Systems and Science, 2nd edition Paul Longley, Michael Goodchild, David Maguire, and David Rhind.
 ISBNs: 0-470-87000-1 (HB); 0-470-87001-X (PB)

Learning Objectives

By the end of this chapter you will:

- Understand the concept of uncertainty, and the ways in which it arises from imperfect representation of geographic phenomena;
- Be aware of the uncertainties introduced in the three stages (conception, measurement and representation, and analysis) of database creation and use;
- Understand the concepts of vagueness and ambiguity, and the uncertainties arising from the definition of key GIS attributes;
- Understand how and why scale of geographic measurement and analysis can both create and propagate uncertainty.

6.1 Introduction

GIS-based representations of the real world are used to reconcile science with practice, concepts with applications, and analytical methods with social context. Yet, almost always, such reconciliation is imperfect, because, necessarily, representations of the world are incomplete (Section 3.4). In this chapter we will use *uncertainty* as an umbrella term to describe the problems that arise out of these imperfections. Occasionally, representations may approach perfect accuracy and precision (terms that we will define in Section 6.3.2.2) – as might be the case, for example, in the detailed site layout layer of a utility management system, in which strenuous efforts are made to reconcile fine-scale multiple measurements of built environments. Yet perfect, or nearly perfect, representations of reality are the exception rather than the rule. More usually, the inherent complexity and detail of our world makes it virtually impossible to capture every single facet, at every possible scale, in a digital representation. (Neither is this usually desirable: see the discussion of sampling in Section 4.4.) Furthermore, different individuals see the world in different ways, and in practice no single view is likely to be accepted universally as the best or to enjoy uncontested status. In this chapter we discuss how the processes and procedures of abstraction create differences between the contents of our (geographic and attribute) database and real-world phenomena. Such differences are almost inevitable and understanding of them can help us to manage uncertainty, and to live with it.

It is impossible to make a perfect representation of the world, so uncertainty about it is inevitable.

Various terms are used to describe differences between the real world and how it appears in a GIS, depending upon the context. The established scientific notion of measurement *error* focuses on differences between observers or between measuring instruments. As we saw in a previous chapter (Section 4.7), the concept of error in multivariate statistics arises in part from omission of some relevant aspects of a phenomenon – as in the failure to fully specify all of the predictor variables in a multiple regression model, for example. Similar problems arise when one or more variables are omitted from the calculation of a composite indicator – as, for example, in omitting road accessibility in an index of land value, or omitting employment status from a measure of social deprivation (see Section 16.2.1 for a discussion of indicators). More generally, the Dutch geostatistician Gerard Heuvelink (who we will introduce in Box 6.1) has defined *accuracy* as the difference between reality and *our* representation of reality. Although such differences might principally be addressed in formal mathematical terms, the use of the word *our* acknowledges the varying views that are generated by a complex, multi-scale, and inherently uncertain world.

Yet even this established framework is too simple for understanding quality or the defining standards of geographic data. The terms *ambiguity* and *vagueness* identify further considerations which need to be taken into account in assessing the *quality* of a GIS representation. Quality is an important topic in GIS, and there have been many attempts to identify its basic dimensions. The US Federal Geographic Data Committee's various standards list five components of quality: attribute accuracy, positional accuracy, logical consistency, completeness, and lineage. Definitions and other details on each of these and several more can be found on the FGDC's Web pages (**www.fgdc.gov**). Error, inaccuracy, ambiguity, and vagueness all contribute to the notion of uncertainty in the broadest sense, and uncertainty may thus be defined as a measure of the user's understanding of the difference between the contents of a dataset, and the real phenomena that the data are believed to represent. This definition implies that phenomena are real, but includes the possibility that we are unable to describe them exactly. In GIS, the term uncertainty has come to be used as the catch-all term to describe situations in which the digital representation is simply incomplete, and as a measure of the general quality of the representation.

Many geographic representations depend upon inherently vague definitions and concepts

The views outlined in the previous paragraph are themselves controversial, and a rich ground for endless philosophical discussions. Some would argue that uncertainty can be inherent in phenomena themselves, rather than just in their description. Others would argue for distinctions between *vagueness, uncertainty, fuzziness, imprecision, inaccuracy*, and many other terms that most people use as if they were essentially synonymous. Information scientist

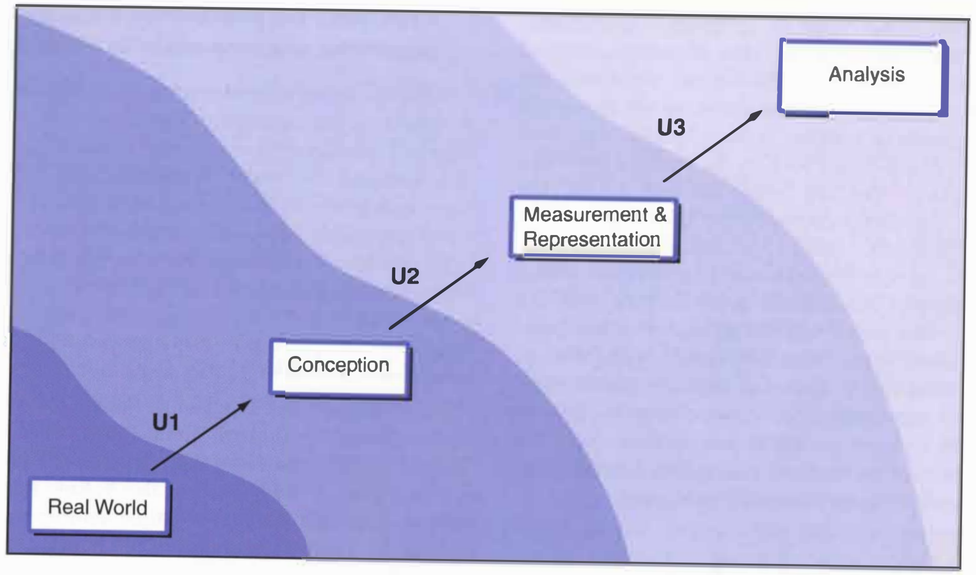


Figure 6.1 A conceptual view of uncertainty. The three filters, U1, U2, and U3 can distort the way in which the complexity of the real world is conceived, measured and represented, and analyzed in a cumulative way

Peter Fisher has provided a useful and wide-ranging discussion of these terms. We take the catch-all view here, and leave the detailed arguments to further study.

In this chapter, we will discuss some of the principal sources of uncertainty and some of the ways in which uncertainty degrades the quality of a spatial representation. The way in which we conceive of a geographic phenomenon very much prescribes the way in which we are likely to set about measuring and representing it. The measurement procedure, in turn, heavily conditions the ways in which it may be analyzed within a GIS. This chain sequence of events, in which *conception* prescribes *measurement and representation*, which in turn prescribes *analysis* is a succinct way of summarizing much of the content of this chapter, and is summarized in Figure 6.1. In this diagram, U1, U2, and U3 each denote *filters* that selectively distort or transform the representation of the real world that is stored and analyzed in GIS: a later chapter (Section 13.2.1) introduces a fourth filter that mediates interpretation of analysis, and the ways in which feedback may be accommodated through improvements in representation.

6.2 U1: Uncertainty in the conception of geographic phenomena

6.2.1 Units of analysis

Our discussion of Tobler's Law (Section 3.1) and of spatial autocorrelation (Section 4.6) established that geographic data handling is different from all other classes of non-spatial applications. A further characteristic that sets geographic information science apart from most every other science is that it is only rarely founded upon *natural* units of analysis. What is the natural unit of measurement for a soil profile? What is the spatial extent of a *pocket* of high unemployment, or a *cluster* of cancer cases? How might we delimit an environmental impact study of spillage from an oil tanker (Figure 6.2)? The questions become still more difficult in bivariate (two variable) and multivariate (more than two variable) studies. At what scale is it appropriate to investigate any relationship between background radiation and the incidence of leukemia? Or to assess any relationship between labor-force qualifications and unemployment rates?

In many cases there are no natural units of geographic analysis.

Figure 6.2 How might the spatial impact of an oil tanker spillage be delineated? We can measure the dispersion of the pollutants, but their impacts extend far beyond these narrowly defined boundaries (Reproduced by permission of Sam C. Pierson, Jr., Photo Researchers)

The discrete object view of geographic phenomena is much more reliant upon the idea of natural units of analysis than the field view. Biological organisms are almost always natural units of analysis, as are groupings such as households or families – though even here there are certainly difficult cases, such as the massive networks of fungal strands that are often claimed to be the largest living organisms on Earth, or extended families of human individuals. Things we manipulate, such as pencils, books, or screwdrivers, are also obvious natural units. The examples listed in the previous paragraph fall almost entirely into one of two categories – they are either instances of fields, where variation can be thought of as inherently continuous in space, or they are instances of poorly defined aggregations of discrete objects. In both of these cases it is up to the investigator to make the decisions about units of analysis, making the identification of the objects of analysis inherently subjective.

6.2.2 Vagueness and ambiguity

6.2.2.1 Vagueness

The frequent absence of objective geographic individual units means that, in practice, the labels that we assign to zones are often vague best guesses. What absolute or relative incidence of oak trees in a forested zone qualifies it for the label *oak woodland* (Figure 6.3)? Or, in a developing-country context in which aerial photography rather than ground enumeration is used to estimate population size, what rate of incidence of dwellings identifies a zone of *dense* population? In each of these instances, it is expedient to transform point-like events (individual trees or individual dwellings) into area objects, and pragmatic decisions must be taken in order to create a working definition of a spatial distribution. These decisions have no absolute validity, and raise two important questions:

- Is the defining boundary of a zone crisp and well-defined?
- Is our assignment of a particular label to a given zone robust and defensible?

Figure 6.3 Seeing the wood for the trees: what absolute or relative incidence rate makes it meaningful to assign the label 'oak woodland'? (Reproduced by permission of Ellan Young, Photo Researchers)

Uncertainty can exist both in the positions of the boundaries of a zone and in its attributes.

The questions have statistical implications (can we put numbers on the confidence associated with boundaries or labels?), cartographic implications (how can we convey the meaning of vague boundaries and labels through appropriate symbols on maps and GIS displays?), and cognitive implications (do people subconsciously attempt to force things into categories and boundaries to satisfy a deep need to simplify the world?).

6.2.2.2 Ambiguity

Many objects are assigned different labels by different national or cultural groups, and such groups perceive space differently. Geographic prepositions like *across*, *over*, and *in* (used in the Yellow Pages query in Figure 1.17) do not have simple correspondences with terms in other languages. Object names and the topological relations between them may thus be inherently *ambiguous*. Perception, behavior, language, and cognition all play a part in the conception of real-world entities and the relationships between them. GIS cannot present a value-neutral view of the world, yet it can provide a formal framework for the reconciliation of different worldviews. The geographic nature of this ambiguity may even be exploited to identify regions with shared characteristics and worldviews. To this end, Box 6.1 describes how different surnames used to describe essentially the same historic occupations provide an enduring measure in region building.

Many linguistic terms used to convey geographic information are inherently ambiguous.

Ambiguity also arises in the conception and construction of *indicators* (see also Section 16.2.1). *Direct* indicators are deemed to bear a clear correspondence with a mapped phenomenon. Detailed household income figures, for example, provide a direct indicator of the likely geography of expenditure and demand for goods and services; tree diameter at breast height can be used to estimate stand value; and field nutrient measures can be used to estimate agronomic yield. *Indirect* indicators are used when the best available measure is a perceived surrogate link with the phenomenon of interest. Thus the incidence of central heating amongst households, or rates of multiple car ownership, might provide a surrogate for (unavailable) household income data, while local atmospheric measurements of nitrogen dioxide might provide an indirect indicator of environmental health. Conception of the (direct or indirect) linkage between any indicator and the phenomenon of interest is subjective, hence ambiguous. Such measures will create (possibly systematic) errors of measurement if the correspondence between the two is imperfect. So, for example, differences in the conception of what hardship and deprivation entail can lead to specification of different composite indicators, and different geodemographic systems include different cocktails of census variables (Section 2.3.3). With regard to the natural environment, conception of critical defining properties

Applications Box 6.1

Historians need maps of our uncertain past

In the study of history, there are many ways in which 'spatial is special' (Section 1.1.1). For example, it is widely recognized that although what our ancestors did (their occupations) and the social groups (classes) to which they belonged were clearly important in terms of demographic behavior, location and place were of equal if not greater importance. Although population changes occur in particular socio-economic circumstances, they are also strongly influenced by the unique characteristics, or 'cultural identities', of particular places. In Great Britain today, as almost everywhere else in the world, most people still think of their nation as made up of 'regions', and their stability and defining characteristics are much debated by cultural geographers and historians.

Yet analyzing and measuring human activity by place creates particular problems for historians. Most obviously, the past was very much less data rich than the present, and few systematic data sources survive. Moreover, the geographical administrative units by which the events of the past were recorded are both complex and changing. In an ideal world, perhaps, physical and cultural boundaries would always coincide, but physical features alone rarely provide appropriate *indicators* of the limits of socio-economic conditions and cultural circumstance.

Unfortunately many mapped historical data are still presented using high-level aggregations, such as counties or regions. This achieves a measure of standardization but may depict demography in only the most arbitrary of ways. If data are forced into geographic administrative units that were delineated for other purposes, regional maps may present nothing more than misleading, or even meaningless, spatial means (see Box 1.9).

In England and in many other countries, the daily activities of most individuals historically revolved around small numbers of contiguous civil *parishes*, of which there were more than 16 000 in the 19th century. These are the smallest administrative units for which data are systematically available. They provide the best available building blocks for meaningful region building. But how can we group parishes in order to identify non-overlapping geographic territories to which people felt that they belonged? And what indicators of regional identity are likely to have survived for all individuals in the population?

Historian Kevin Schürer (Box 13.2) has investigated these questions using a historical GIS to map digital surname data from the 1881 Census of England and Wales. The motivation for the GIS arises from the observation that many surnames contain statements of regional identity, and the suggestion that distinct zones of similar surnames might be described as homogeneous regions. The digitized records of the 1881 Census for England and Wales cover some 26 million people: although some 41 000 different surnames are recorded, a fifth of the population shared just under 60 surnames, and half of the population were accounted for by some 600 surnames. Schürer suggests that these aggregate statistics conceal much that we might learn about regional identity and diversity.

Many surnames of European origin are formed from occupational titles. Occupations often have uneven regional distributions and sometimes similar occupations are described using different names in different places (at the global scale, today's 'realtors' in the US perform much the same functions as their 'estate agent' counterparts in the UK, for example). Schürer has investigated the 1881 geographical distribution of three occupational surnames – Fuller, Tucker, and Walker. These essentially refer to the same occupation; namely someone who, from around the 14th century onwards, worked in the preparation of textiles by scouring or beating cloth as a means of finishing or cleansing it. Using GIS, Schürer confirms that the geographies of these 14th century surnames remained of enduring importance in defining the regional geography of England in 1881. Figure 6.4 illustrates that in 1881 Tuckers remained concentrated in the West Country, while Fullers occurred principally in the east and Walkers resided in the Midlands and north. This map also shows that there was not much mixing of the surnames in the transition zones between names, suggesting that the maps provide a useful basis to region building.

The enduring importance of surnames as evidence of the strength and durability of regional cultures has been confirmed in an update to the work by Daryl Lloyd at University College London: Lloyd used the 2003 UK Electoral Register to map the distribution of the same three surnames (Figure 6.5) and identified persistent regional concentrations.

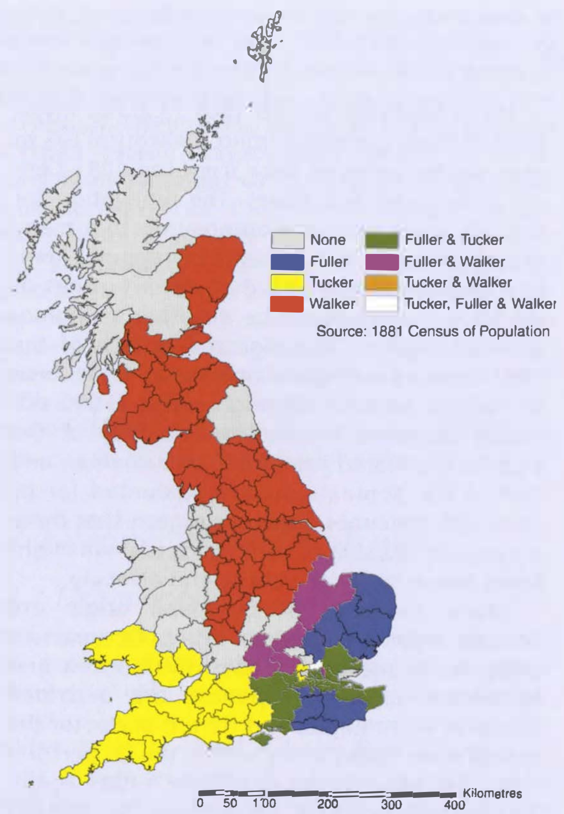


Figure 6.4 The 1881 geography of the Fullers, Tuckers, and Walkers (Reproduced with permission of K. Schürer)

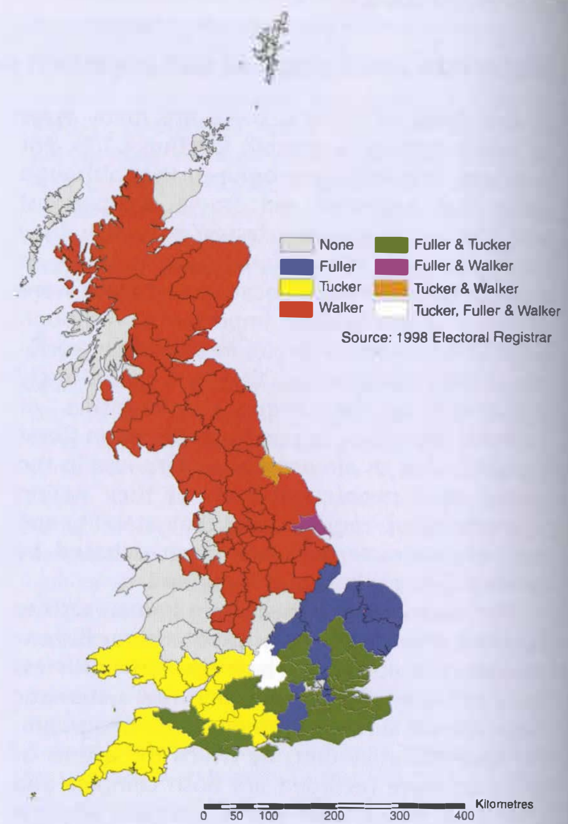


Figure 6.5 The 2003 geography of the Fullers, Tuckers, and Walkers (Reproduced with permission of Daryl Lloyd)

of soils can lead to inherent ambiguity in their classification (see Section 6.2.4).

Ambiguity is introduced when imperfect indicators of phenomena are used instead of the phenomena themselves.

Fundamentally, GIS has upgraded our abilities to generalize about spatial distributions. Yet our abilities to do so may be constrained by the different taxonomies that are conceived and used by data-collecting organizations within our overall study area. A study of wetland classification in the US found no fewer than six agencies engaged in mapping the same phenomena over the same geographic areas, and each with their own definitions of wetland types (see Section 1.2). If wetland maps are to be used in regulating the use of land, as they are in many areas, then uncertainty in mapping clearly exposes regulatory agencies to potentially damaging and costly lawsuits. How might soils data classified according to the UK national classification be assimilated within a pan-European soils map, which uses a classification honed to the full range and diversity of soils found across the European continent rather than those just on an offshore island? How might different national geodemographic classifications be combined into a form suitable for a pan-European marketing exercise? These are all variants of the question:

- How may mismatches between the categories of different classification schema be reconciled?

Differences in definitions are a major impediment to integration of geographic data over wide areas.

Like the process of pinning down the different nomenclatures developed in different cultural settings, the process of reconciling the semantics of different classification schema is an inherently *ambiguous* procedure. Ambiguity arises in data concatenation when we are unsure regarding the *meta-category* to which a particular class should be assigned.

6.2.3 Fuzzy approaches

One way of resolving the assignment process is to adopt a probabilistic interpretation. If we take a statement like 'the database indicates that this field contains wheat, but there is a 0.17 probability (or 17% chance) that it actually contains barley', there are at least two possible interpretations:

(a) If 100 randomly chosen people were asked to make independent assessments of the field on the ground, 17 would determine that it contains barley, and 83 would decide it contains wheat.

(b) Of 100 similar fields in the database, 17 actually contained barley when checked on the ground, and 83 contained wheat.

Of the two we probably find the second more acceptable because the first implies that people cannot correctly determine the crop in the field. But the important point is that, in conceptual terms, both of these interpretations are *frequentist*, because they are based on the notion that the probability of a given outcome can be defined as the proportion of times the outcome occurs in some real or imagined experiment, when the number of tests is very large. Yet while this is reasonable for classic statistical experiments, like tossing coins or drawing balls from an urn, the geographic situation is different – there is only one field with precisely these characteristics, and one observer, and in order to imagine a number of tests we have to invent more than one observer, or more than one field (the problems of imagining larger populations for some geographic samples are discussed further in Section 15.4).

In part because of this problem, many people prefer the *subjectivist* conception of probability – that it represents a judgment about relative likelihood that is not the result of any frequentist experiment, real or imagined. Subjective probability is similar in many ways to the concept of fuzzy sets, and the latter framework will be used here to emphasize the contrast with frequentist probability.

Suppose we are asked to examine an aerial photograph to determine whether a field contains wheat, and we decide that we are not sure. However, we are able to put a number on our degree of uncertainty, by putting it on a scale from 0 to 1. The more certain we are, the higher the number. Thus we might say we are 0.90 sure it is wheat, and this would reflect a greater degree of certainty than 0.80. This degree of belonging to the class *wheat* is termed the *fuzzy membership*, and it is common though not necessary to limit memberships to the range 0 to 1. In effect, we have changed our view of membership in classes, and abandoned the notion that things must either belong to classes or not belong to them – in this new world, the boundaries of classes are no longer clean and crisp, and the set of things assigned to a set can be fuzzy.

In fuzzy logic, an object's degree of belonging to a class can be partial.

One of the major attractions of fuzzy sets is that they appear to let us deal with sets that are not precisely defined, and for which it is impossible to establish membership cleanly. Many such sets or classes are found in GIS applications, including land use categories, soil types, land cover classes, and vegetation types. Classes used for maps are often fuzzy, such that two people asked to classify the same location might disagree, not because of measurement error, but because the classes themselves are not perfectly defined and because opinions vary. As such, mapping is often forced to stretch the rules of scientific repeatability, which require that two observers will always agree. Box 6.2 shows a typical extract from the legend of a soil map, and it is easy to see how two people might disagree, even though both are experts with years of experience in soil classification.

Figure 6.6 shows an example of mapping classes using the fuzzy methods developed by A-Xing Zhu of the

Technical Box 6.2

Fuzziness in classification: description of a soil class

Following is the description of the Limerick series of soils from New England, USA (the type location is in Chittenden County, Vermont), as defined by the National Cooperative Soil Survey. Note the frequent use of vague terms such as 'very', 'moderate', 'about', 'typically', and 'some'. Because the definition is so loose it is possible for many distinct soils to be lumped together in this one class – and two observers may easily disagree over whether a given soil belongs to the class, even though both are experts. The definition illustrates the extreme problems of defining soil classes with sufficient rigor to satisfy the criterion of scientific repeatability.

The Limerick series consists of very deep, poorly drained soils on flood plains. They formed in loamy alluvium. Permeability is moderate. Slope ranges from 0 to 3 percent. Mean annual precipitation is about 34 inches and mean annual temperature is about 45 degrees F.

Depth to bedrock is more than 60 inches. Reaction ranges from strongly acid to neutral in the surface layer and moderately acid to neutral in the substratum. Textures are typically silt loam or very fine sandy loam, but lenses of loamy very fine sand or very fine sand are present in some pedons. The weighted average of fine and coarser sands, in the particle-size control section, is less than 15 percent.

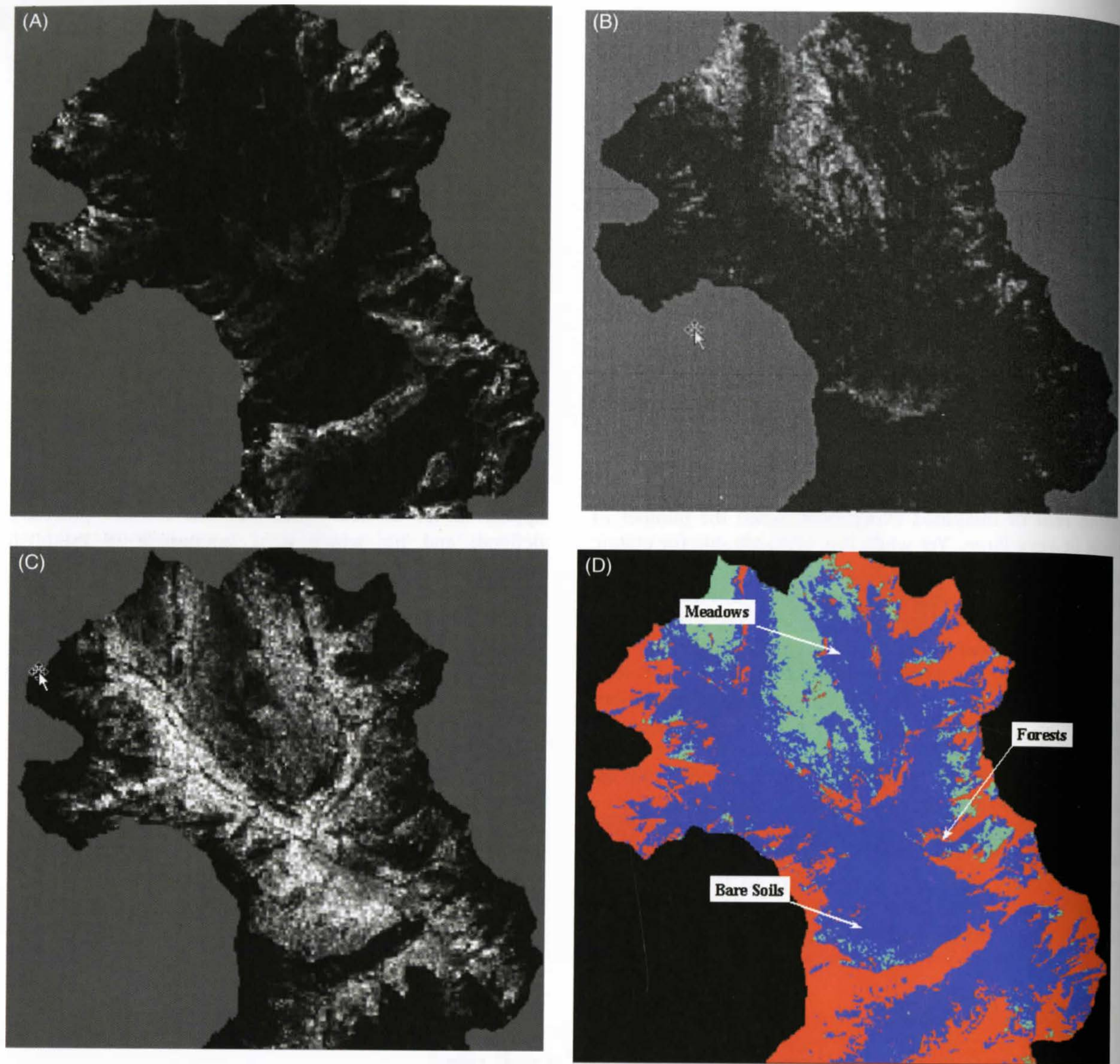


Figure 6.6 (A) Membership map for bare soils in the Upper Lake McDonald basin, Glacier National Park. High membership values are in the ridge areas where active colluvial and glacier activities prevent the establishment of vegetation. (B) Membership map for forest. High membership values are in the middle to lower slope areas where the soils are both stable and better drained.
(C) Membership map for alpine meadows. High membership values are on gentle slopes at high elevation where excessive soil water and low temperature prevent the growth of trees. (D) Spatial distribution of the three cover types from hardening the membership maps. (Reproduced by permission of A-Xing Zhu)

University of Wisconsin-Madison, USA, which take both remote sensing images and the opinions of experts as inputs. There are three classes, and each map shows the fuzzy membership values in one class, ranging from 0 (darkest) to 1 (lightest). This figure also shows the result of converting to *crisp* categories, or *hardening* – to obtain Figure 6.6D, each pixel is colored according to the class with the highest membership value.

Fuzzy approaches are attractive because they capture the uncertainty that many of us feel about the assignment of places on the ground to specific categories. But researchers have struggled with the question of whether they are more *accurate*. In a sense, if we are uncertain about which class to choose then it is more accurate to say so, in the form of a fuzzy membership, than to be forced into assigning a class without qualification. But that does not address the question of whether the fuzzy membership value is accurate. If Class A is not well defined, it is hard to see how one person's assignment of a fuzzy membership of 0.83 in Class A can be meaningful to another person, since there is no reason to believe that the two people share the same notions of what Class A means, or of what 0.83 means, as distinct from 0.91, or 0.74. So while fuzzy approaches make sense at an intuitive level, it is more difficult to see how they could be helpful in the

process of communication of geographic knowledge from one person to another.

6.2.4 The scale of geographic individuals

There is a sense in which vagueness and ambiguity in the conception of *usable* (rather than *natural*) units of analysis undermines the very foundations of GIS. How, in practice, may we create a sufficiently secure base to support geographic analysis? Geographers have long grappled with the problems of defining systems of zones and have marshaled a range of deductive and inductive approaches to this end (see Section 4.9 for a discussion of what deduction and induction entail). The long-established regional geography tradition is fundamentally concerned with the delineation of zones characterized by internal homogeneity (with respect to climate, economic development, or agricultural land use, for example), within a zonal scheme which maximizes between-zone heterogeneity, such as the map illustrated in Figure 6.7. Regional geography is fundamentally about delineating *uniform* zones, and many employ multivariate statistical techniques such as cluster analysis to supplement, or post-rationalize, intuition.

Identification of homogeneous zones and spheres of influence lies at the heart of traditional regional geography as well as contemporary data analysis.

Other geographers have tried to develop *functional* zonal schemes, in which zone boundaries delineate the breakpoints between the spheres of influence of adjacent facilities or features – as in the definition of travel-to-work areas (Figure 6.8) or the definition of a river catchment. Zones may be defined such that there is maximal interaction within zones, and minimal between zones. The scale at which uniformity or functional integrity is conceived clearly conditions the ways it is measured – in terms of the magnitude of within-zone heterogeneity that must be accommodated in the case of uniform zones, and the degree of leakage between the units of functional zones.

Scale has an effect, through the concept of spatial autocorrelation outlined in Section 4.3, upon the outcome of geographic analysis. This was demonstrated more than half a century ago in a classic paper by Yule and Kendall, where the correlation between wheat and potato yields was shown systematically to increase as English county units were amalgamated through a succession of coarser scales (Table 6.1). A succession of research papers has subsequently reaffirmed the existence of similar scale effects in multivariate analysis. However, rather discouragingly, scale effects in multivariate cases do not follow any consistent or predictable trends. This theme of dependence of results on the geographic units of analysis is pursued further in Section 6.4.3.

Relationships typically grow stronger when based on larger geographic units.

GIS appears to trivialize the task of creating composite thematic maps. Yet inappropriate conception of the scale of geographic phenomena can mean that apparent spatial

Figure 6.7 The regional geography of Russia. (*Source*: de Blij H.J. and Muller P.O. 2000 *Geography: Realms, Regions and Concepts* (9th edn) New York: Wiley, p. 113)

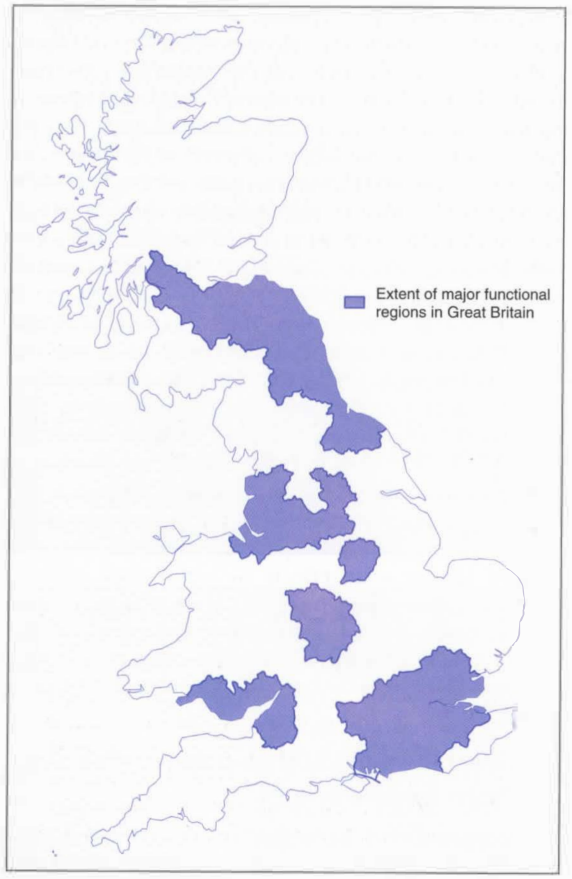


Figure 6.8 Dominant functional regions of Great Britain. (*Source*: Champion A.G., Green A.E., Owen D.W., Ellin D.J., Coombes M.G. 1987 *Changing Places: Britain's Demographic, Economic and Social Complexion*, London: Arnold, p. 9)

Table 6.1 In 1950 Yule and Kendall used data for wheat and potato yields from the (then) 48 counties of England to demonstrate that correlation coefficients tend to increase with scale. They aggregated the 48-county data into zones so that there were first 24, then 12, then 6, and finally just 3 zones. The range of their results, from near zero (no correlation) to over 0.99 (almost perfect positive correlation) demonstrates the range of results that can be obtained, although subsequent research has suggested that this range of values is atypical

No. of geographic areas	Correlation
48	0.2189
24	0.2963
12	0.5757
6	0.7649
3	0.9902

patterning (or the lack of it) in mapped data may be oversimplified, crude, or even illusory. It is also clearly inappropriate to conceive of boundaries as crisp and well-defined if significant leakage occurs across them (as happens, in practice, in the delineation of most functional regions), or if geographic phenomena are by nature fuzzy, vague, or ambiguous.

6.3 U2: Further uncertainty in the measurement and representation of geographic phenomena

6.3.1 Measurement and representation

The conceptual models (fields and objects) that were introduced in Chapter 3 impose very different filters upon reality, and their usual corresponding representational models (raster and vector) are characterized by different uncertainties as a consequence. The vector model enables a range of powerful analytical operations to be performed (see Chapters 14 through 16), yet it also requires *a priori* conceptualization of the nature and extent of geographic individuals and the ways in which they nest together into higher-order zones. The raster model defines individual elements as square cells, with boundaries that bear no relationship at all to natural features, but nevertheless provides a convenient and (usually) efficient structure for data handling within a GIS. However, in the absence of effective automated pattern recognition techniques, human interpretation is usually required to discriminate between real-world spatial entities as they appear in a rasterized image.

Although quite different representations of reality, vector and raster data structures are both attractive in their logical consistency, the ease with which they are able to handle spatial data, and (once the software is written) the ease with which they can be implemented in GIS. But neither abstraction provides easy measurement fixes and there is no substitute for robust conception of geographic units of analysis (Section 6.2). This said, however, the conceptual distinction between fields and discrete objects is often useful in dealing with uncertainty. Figure 6.9 shows a coastline, which is often conceptualized as a discrete line object. But suppose we recognize that its position is uncertain. For example, the coastline shown on a 1:2 000 000 map is a gross generalization, in which major liberties are taken, particularly in areas where the coast is highly indented and irregular. Consequently the 1:2 000 000 version leaves substantial uncertainty about the true location of the shoreline. We might approach this by changing from a line to an area, and mapping the area where the actual coastline lies, as shown in the figure. But another approach would be to reconceptualize the coastline as a field, by mapping a variable whose value represents the probability that a point is land. This is shown in the figure as a raster representation. This would have far more information content, and consequently much more value in many applications. But at the same time it would be difficult to find an appropriate data source for the representation – perhaps a fuzzy classification of an air photo, using one of an increasing number of techniques designed to produce representations of the uncertainty associated with objects discovered in images.

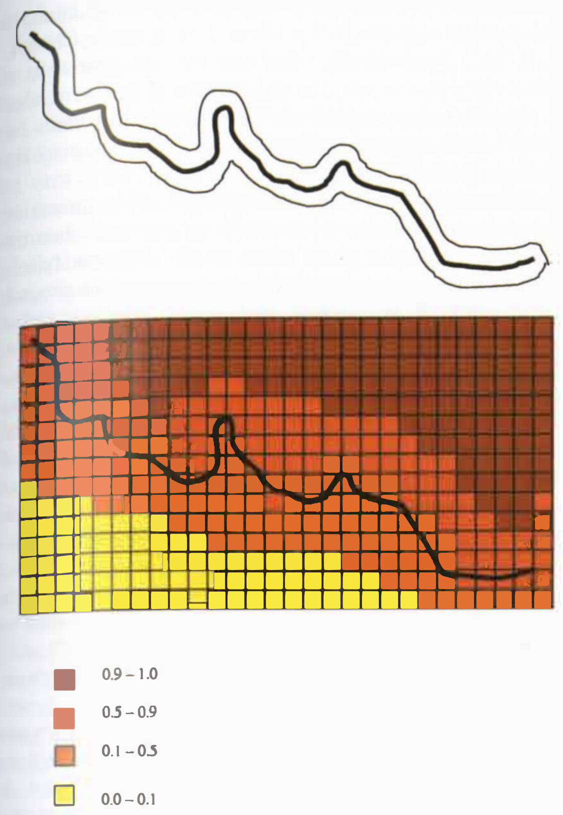


Figure 6.9 The contrast between discrete object (top) and field (bottom) conceptualizations of an uncertain coastline. In the discrete object view the line becomes an area delimiting where the true coastline might be. In the field view a continuous surface defines the probability that any point is land

Uncertainty can be measured differently under field and discrete object views.

Indeed, far from offering quick fixes for eliminating or reducing uncertainty, the measurement process can actually increase it. Given that the vector and raster data models impose quite different filters on reality, it is unsurprising that they can each generate additional uncertainty in rather different ways. In field-based conceptualizations, such as those that underlie remotely sensed images expressed as rasters, spatial objects are not defined *a priori*. Instead, the classification of each cell into one or other category builds together into a representation. In remote sensing, when resolution is insufficient to detect all of the detail in geographic phenomena, the term *mixel* is often used to describe raster cells that contain more than one class of land – in other words, elements in which the outcome of statistical classification suggests the occurrence of multiple land cover categories. The total area of cells classified as mixed should decrease as the resolution of the satellite sensor increases, assuming the number of categories remains constant, yet a completely mixel-free classification is very unlikely at any level of resolution. Even where the Earth's surface is covered with perfectly homogeneous areas, such as agricultural fields growing uniform crops, the failure of real-world crop boundaries to line up with pixel edges ensures the presence of at least some mixels. Neither does higher-resolution imagery solve all problems: medium-resolution data (defined as pixel size of between 30 m × 30 m and 1000 m × 1000 m) are typically classified using between 3 and 7 bands, while high-resolution data (pixel sizes 10 × 10 m or smaller) are typically classified using between 7 and 256 bands, and this can generate much greater heterogeneity of spectral values with attendant problems for classification algorithms.

A pixel whose area is divided among more than one class is termed a mixel.

The vector data structure, by contrast, defines spatial entities and specifies explicit topological relations (see Section 3.6) between them. Yet this often entails transformations of the inherent characteristics of spatial objects (Section 14.4). In conceptual terms, for example, while the true individual members of a population might each be defined as point-like objects, they will often appear in a GIS dataset only as aggregate counts for apparently *uniform* zones. Such aggregation can be driven by the need to preserve confidentiality of individual records, or simply by the need to limit data volume. Unlike the field conceptualization of spatial phenomena, this implies that there are good reasons for partitioning space in a particular way. In practice, partitioning of space is often made on grounds that are principally pragmatic, yet are rarely completely random (see Section 6.4). In much of socio-economic GIS, for example, zones which are designed to preserve the anonymity of survey respondents may be largely *ad hoc* containers. Larger aggregations are often used for the simple reason that they permit comparisons of measures over time (see Box 6.1). They may also reflect the way that a cartographer or GIS interpolates a boundary between sampled points, as in the creation of isopleth maps (Box 4.3).

6.3.2 Statistical models of uncertainty

Scientists have developed many widely used methods for describing errors in observations and measurements, and these methods may be applicable to GIS if we are willing to think of databases as collections of measurements. For example, a digital elevation model consists of a large number of measurements of the elevation of the Earth's surface. A map of land use is also in a sense a collection of measurements, because observations of the land surface have resulted in the assignment of classes to locations. Both of these are examples of observed or measured attributes, but we can also think of location as a property that is measured.

A geographic database is a collection of measurements of phenomena on or near the Earth's surface.

Here we consider errors in nominal class assignment, such as of types of land use, and errors in continuous (interval or ratio) scales, such as elevation (see Section 3.4).

6.3.2.1 Nominal case

The values of nominal data serve only to distinguish an instance of one class from an instance of another, or to identify an object uniquely. If classes have an inherent ranking they are described as ordinal data, but for purposes of simplicity the ordinal case will be treated here as if it were nominal.

Consider a single observation of nominal data – for example, the observation that a single parcel of land is being used for agriculture (this might be designated by giving the parcel Class A as its value of the 'Land Use Class' attribute). For some reason, perhaps related to the quality of the aerial photography being used to build the database, the class may have been recorded falsely as Class G, Grassland. A certain proportion of parcels that are truly Agriculture might be similarly recorded as Grassland, and we can think of this in terms of a probability, that parcels that are truly Agriculture are falsely recorded as Grassland.

Table 6.2 shows how this might work for all of the parcels in a database. Each parcel has a true class, defined by accurate observation in the field, and a recorded class as it appears in the database. The whole table is described as a *confusion matrix*, and instances of confusion matrices are commonly encountered in applications dominated by class data, such as classifications derived from remote sensing or aerial photography. The true class might be determined by ground check, which is inherently more accurate than classification of aerial photographs, but much more expensive and time-consuming.

Ideally all of the observations in the confusion matrix should be on the principal diagonal, in the cells that correspond to agreement between true class and database class. But in practice certain classes are more easily confused than others, so certain cells off the diagonal will have substantial numbers of entries.

A useful way to think of the confusion matrix is as a set of rows, each defining a vector of values.

Table 6.2 Example of a misclassification or confusion matrix. A grand total of 304 parcels have been checked. The rows of the table correspond to the land use class of each parcel as recorded in the database, and the columns to the class as recorded in the field. The numbers appearing on the principal diagonal of the table (from top left to bottom right) reflect correct classification

	A	B	C	D	E	Total
A	80	4	0	15	7	106
B	2	17	0	9	2	30
C	12	5	9	4	8	38
D	7	8	0	65	0	80
E	3	2	1	6	38	50
Total	104	36	10	99	55	304

The vector for row i gives the proportions of cases in which what appears to be Class i is actually Class 1, 2, 3, etc. Symbolically, this can be represented as a vector $\{p_1, p_2, \ldots, p_i, \ldots, p_n\}$, where n is the number of classes, and p_i represents the proportion of cases for which what appears to be the class according to the database is actually Class i.

There are several ways of describing and summarizing the confusion matrix. If we focus on one row, then the table shows how a given class in the database falsely records what are actually different classes on the ground. For example, Row A shows that of 106 parcels recorded as Class A in the database, 80 were confirmed as Class A in the field, but 15 appeared to be truly Class D. The proportion of instances in the diagonal entries represents the proportion of correctly classified parcels, and the total of off-diagonal entries in the row is the proportion of entries in the database that appear to be of the row's class but are actually incorrectly classified. For example, there were only 9 instances of agreement between the database and the field in the case of Class D. If we look at the table's columns, the entries record the ways in which parcels that are truly of that class are actually recorded in the database. For example, of the 10 instances of Class C found in the field, 9 were recorded as such in the database and only 1 was misrecorded as Class E.

The columns have been called the *producer's* perspective, because the task of the producer of an accurate database is to minimize entries outside the diagonal cell in a given column, and the rows have been called the *consumer's* perspective, because they record what the contents of the database actually mean on the ground; in other words, the accuracy of the database's contents.

Users and producers of data look at misclassification in distinct ways.

For the table as a whole, the proportion of entries in diagonal cells is called the *percent correctly classified* (PCC), and is one possible way of summarizing the table. In this case 209/304 cases are on the diagonal, for a PCC of 68.8%. But this measure is misleading for at least two reasons. First, chance alone would produce some correct classifications, even in the worst circumstances, so it would be more meaningful if the scale were adjusted such that 0 represents chance. In this case, the number of chance hits on the diagonal in a random assignment is 76.2 (the sum of the row total times the column total divided by the grand total for each of the five diagonal cells). So the actual number of diagonal hits, 209, should be compared to this number, not 0. The more useful index of success is the *kappa index*, defined as:

$$\kappa = \frac{\sum_{i=1}^{n} c_{ii} - \sum_{i=1}^{n} c_{i.}c_{.i}/c_{..}}{c_{..} - \sum_{i=1}^{n} c_{i.}c_{.i}/c_{..}}$$

where c_{ij} denotes the entry in row i column j, the dots indicate summation (e.g., $c_{i.}$ is the summation over all columns for row i, that is, the row i total, and $c_{..}$ is the grand total), and n is the number of classes. The first term in the numerator is the sum of all the diagonal entries (entries for which the row number and the column number are the same). To compute PCC we would simply divide this term by the grand total (the first term in the denominator). For kappa, both numerator and denominator are reduced by the same amount, an estimate of the number of hits (agreements between field and database) that would occur by chance. This involves taking each diagonal cell, multiplying the row total by the column total, and dividing by the grand total. The result is summed for each diagonal cell. In this case kappa evaluates to 58.3%, a much less optimistic assessment than PCC.

The second issue with both of these measures concerns the relative abundance of different classes. In the table, Class C is much less common than Class A. The confusion matrix is a useful way of summarizing the characteristics of nominal data, but to build it there must be some source of more accurate data. Commonly this is obtained by ground observation, and in practice the confusion matrix is created by taking samples of more accurate data, by sending observers into the field to conduct spot checks. Clearly it makes no sense to visit every parcel, and instead a sample is taken. Because some classes are commoner than others, a random sample that made every parcel equally likely to be chosen would be inefficient, because too many data would be gathered on common classes, and not enough on the relatively rare ones. So, instead, samples are usually chosen such that a roughly equal number of parcels are selected in each class. Of course these decisions must be based on the class as recorded in the database, rather than the true class. This is an instance of sampling that is systematically *stratified* by class (see Section 4.4).

Sampling for accuracy assessment should pay greater attention to the classes that are rarer on the ground.

Parcels represent a relatively easy case, if it is reasonable to assume that the land use class of a parcel is uniform over the parcel, and class is recorded as a single attribute of each parcel object. But as we noted in Section 6.2, more difficult cases arise in sampling natural areas (for example in the case of vegetation cover class), where parcel boundaries do not exist. Figure 6.10 shows a typical vegetation cover class map, and is obviously highly generalized. If we were to apply the previous strategy, then we would test each area to see if its assigned vegetation cover class checks out on the ground. But unlike the parcel case, in this example the boundaries between areas are not fixed, but are themselves part of the observation process, and we need to ask whether they are correctly located. Error in this case has two forms: misallocation of an area's class and mislocation of an area's boundaries. In some cases the boundary between two areas may be fixed, because it coincides with a clearly defined line on the ground; but in other cases, the boundary's location is as much a matter of judgment as the allocation of an area's class. Peter Burrough and Andrew Frank have discussed many of the implications of uncertain boundaries in GIS.

Figure 6.10 An example of a vegetation cover map. Two strategies for accuracy assessment are available: to check by area (polygon), or to check by point. In the former case a strategy would be devised for field checking each area, to determine the area's correct class. In the latter, points would be sampled across the state and the correct class determined at each point

Errors in land cover maps can occur in the locations of boundaries of areas, as well as in the classification of areas.

In such cases we need a different strategy, that captures the influence both of mislocated boundaries and of misallocated classes. One way to deal with this is to think of error not in terms of classes assigned to areas, but in terms of classes assigned to points. In a raster dataset, the cells of the raster are a reasonable substitute for individual points. Instead of asking whether area classes are confused, and estimating errors by sampling areas, we ask whether the classes assigned to raster cells are confused, and define the confusion matrix in terms of misclassified cells. This is often called *per-pixel* or *per-point* accuracy assessment, to distinguish it from the previous strategy of *per-polygon* accuracy assessment. As before, we would want to stratify by class, to make sure that relatively rare classes were sampled in the assessment.

6.3.2.2 Interval/ratio case

The second case addresses measurements that are made on interval or ratio scales. Here, error is best thought of not as a change of class, but as a change of value, such that the observed value x' is equal to the true value

x plus some distortion δx, where δx is hopefully small. δx might be either positive or negative, since errors are possible in both directions. For example, the measured and recorded elevation at some point might be equal to the true elevation, distorted by some small amount. If the average distortion is zero, so that positive and negative errors balance out, the observed values are said to be *unbiased*, and the average value will be true.

Error in measurement can produce a change of class, or a change of value, depending on the type of measurement.

Sometimes it is helpful to distinguish between *accuracy*, which has to do with the magnitude of δx, and *precision*. Unfortunately there are several ways of defining precision in this context, at least two of which are regularly encountered in GIS. Surveyors and others concerned with measuring instruments tend to define precision through the performance of an instrument in making repeated measurements of the same phenomenon. A measuring instrument is precise according to this definition if it repeatedly gives similar measurements, whether or not these are actually accurate. So a GPS receiver might make successive measurements of the same elevation, and if these are similar the instrument is said to be precise. Precision in this case can be measured by the variability among repeated measurements. But it is possible, for example, that all of the measurements are approximately 5 m too high, in which case the measurements are said to be biased, even though they are precise, and the instrument is said to be inaccurate. Figure 6.11 illustrates this meaning of precise, and its relationship to accuracy.

The other definition of precision is more common in science generally. It defines precision as the number of digits used to report a measurement, and again it is not necessarily related to accuracy. For example, a GPS receiver might measure elevation as 51.3456 m. But if the receiver is in reality only accurate to the nearest 10 cm, three of those digits are spurious, with no real meaning. So, although the precision is one ten thousandth of a meter, the accuracy is only one tenth of a meter. Box 6.3 summarizes the rules that are used to ensure that reported measurements do not mislead by appearing to have greater accuracy than they really do.

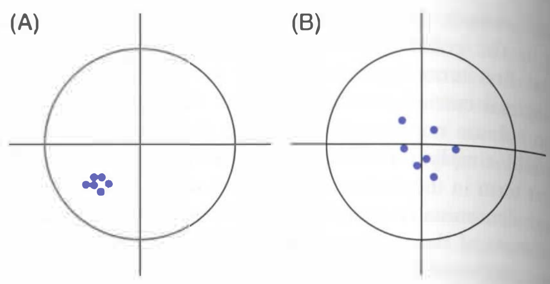


Figure 6.11 The term *precision* is often used to refer to the repeatability of measurements. In both diagrams six measurements have been taken of the same position, represented by the center of the circle. In (A) successive measurements have similar values (they are *precise*), but show a bias away from the correct value (they are *inaccurate*). In (B), precision is lower but accuracy is higher

To most scientists, precision refers to the number of significant digits used to report a measurement, but it can also refer to a measurement's repeatability.

In the interval/ratio case, the magnitude of errors is described by the *root mean square error* (RMSE), defined as the square root of the average squared error, or:

$$\left[\sum \delta x^2/n\right]^{1/2}$$

Technical Box 6.3

Good practice in reporting measurements

Here are some simple rules that help to ensure that people receiving measurements from others are not misled by their apparently high precision.

1. The number of digits used to report a measurement should reflect the measurement's accuracy. For example, if a measurement is accurate to 1 m then no decimal places should be reported. The measurement 14.4 m suggests accuracy to one tenth of a meter, as does 14.0, but 14 suggests accuracy to 1 m.
2. Excess digits should be removed by rounding. Fractions above one half should be rounded up, fractions below one half should be rounded down. The following examples reflect rounding to two decimal places:

   14.57803 rounds to 14.58
   14.57397 rounds to 14.57
   14.57999 rounds to 14.58
   14.57499 rounds to 14.57
3. These rules are not effective to the left of the decimal place – for example, they give no basis for knowing whether 1400 is accurate to the nearest unit, or to the nearest hundred units.
4. If a number is known to be exactly an integer or whole number, then it is shown with no decimal point.

where the summation is over the values of δx for all of the n observations. The RMSE is similar in a number of ways to the standard deviation of observations in a sample. Although RMSE involves taking the square root of the average squared error, it is convenient to think of it as approximately equal to the average error in each observation, whether the error is positive or negative. The US Geological Survey uses RMSE as its primary measure of the accuracy of elevations in digital elevation models, and published values range up to 7 m.

Although the RMSE can be thought of as capturing the magnitude of the average error, many errors will be greater than the RMSE, and many will be less. It is useful, therefore, to know how errors are *distributed* in magnitude – how many are large, how many are small. Statisticians have developed a series of models of error distributions, of which the commonest and most important is the Gaussian distribution, otherwise known as the error function, the 'bell curve', or the Normal distribution. Figure 6.12 shows the curve's shape. If observations are unbiased, then the mean error is zero (positive and negative errors cancel each other out), and the RMSE is also the distance from the center of the distribution (zero) to the points of inflection on either side, as shown in the figure. Let us take the example of a 7 m RMSE on elevations in a USGS digital elevation model; if error follows the Gaussian distribution, this means that some errors will be more than 7 m in magnitude, some will be less, and also that the relative abundance of errors of any given size is described by the curve shown. 68% of errors will be between −1.0 and +1.0 RMSEs, or −7 m and +7 m. In practice many distributions of error do follow the Gaussian distribution, and there are good theoretical reasons why this should be so.

The Gaussian distribution predicts the relative abundances of different magnitudes of error.

To emphasize the mathematical formality of the Gaussian distribution, its equation is shown below. The symbol σ denotes the standard deviation, μ denotes the mean (in Figure 6.12 these values are 1 and 0 respectively), and exp is the exponential function, or '2.71828 to the power of'. Scientists believe that it applies very broadly, and that many instances of measurement error adhere closely to the distribution, because it is grounded in rigorous theory. It can be shown mathematically that the distribution arises whenever a large number of random factors contribute to error, and the effects of these factors combine additively – that is, a given effect makes the same additive contribution to error whatever the specific values of the other factors. For example, error might be introduced in the use of a steel tape measure over a large number of measurements because some observers consistently pull the tape very taught, or hold it very straight, or fastidiously keep it horizontal, or keep it cool, and others do not. If the combined effects of these considerations always contributes the same amount of error (e.g., +1 cm, or −2 cm), then this contribution to error is said to be additive.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

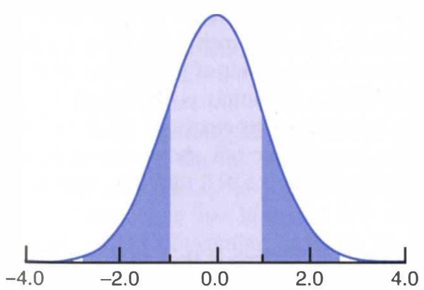


Figure 6.12 The Gaussian or Normal distribution. The height of the curve at any value of x gives the relative abundance of observations with that value of x. The area under the curve between any two values of x gives the probability that observations will fall in that range. The range between −1 standard deviation and +1 standard deviation is in light purple. It encloses 68% of the area under the curve, indicating that 68% of observations will fall between these limits

We can apply this idea to determine the inherent uncertainty in the locations of contours. The US Geological Survey routinely evaluates the accuracies of its digital elevation models (DEMs), by comparing the elevations recorded in the database with those at the same locations in more accurate sources, for a sample of points. The differences are summarized in a RMSE, and in this example we will assume that errors have a Gaussian distribution with zero mean and a 7 m RMSE. Consider a measurement of 350 m. According to the error model, the truth might be as high as 360 m, or as low as 340 m, and the relative frequencies of any particular error value are as predicted by the Gaussian distribution with a mean of zero and a standard deviation of 7. If we take error into account, using the Gaussian distribution with an RMSE of 7 m, it is no longer clear that a measurement of 350 m lies exactly on the 350 m contour. Instead, the truth might be 340 m, or 360 m, or 355 m. Figure 6.13 shows the implications of this in terms of the location of this contour in a real-world example. 95% of errors would put the contour within the colored zone. In areas colored red the observed value is less than 350 m, but the truth might be 350 m; in areas colored green the observed value is more than 350 m, but the truth might be 350 m. There is a 5% chance that the true location of the contour lies outside the colored zone entirely.

6.3.3 Positional error

In the case of measurements of position, it is possible for every coordinate to be subject to error. In the two-dimensional case, a measured position (x', y') would be subject to errors in both x and y; specifically, we might write $x' = x + \delta x$, $y' = y + \delta y$, and similarly in the three-dimensional case where all three coordinates are measured, $z' = z + \delta z$. The *bivariate Gaussian distribution* describes errors in the two horizontal dimensions,

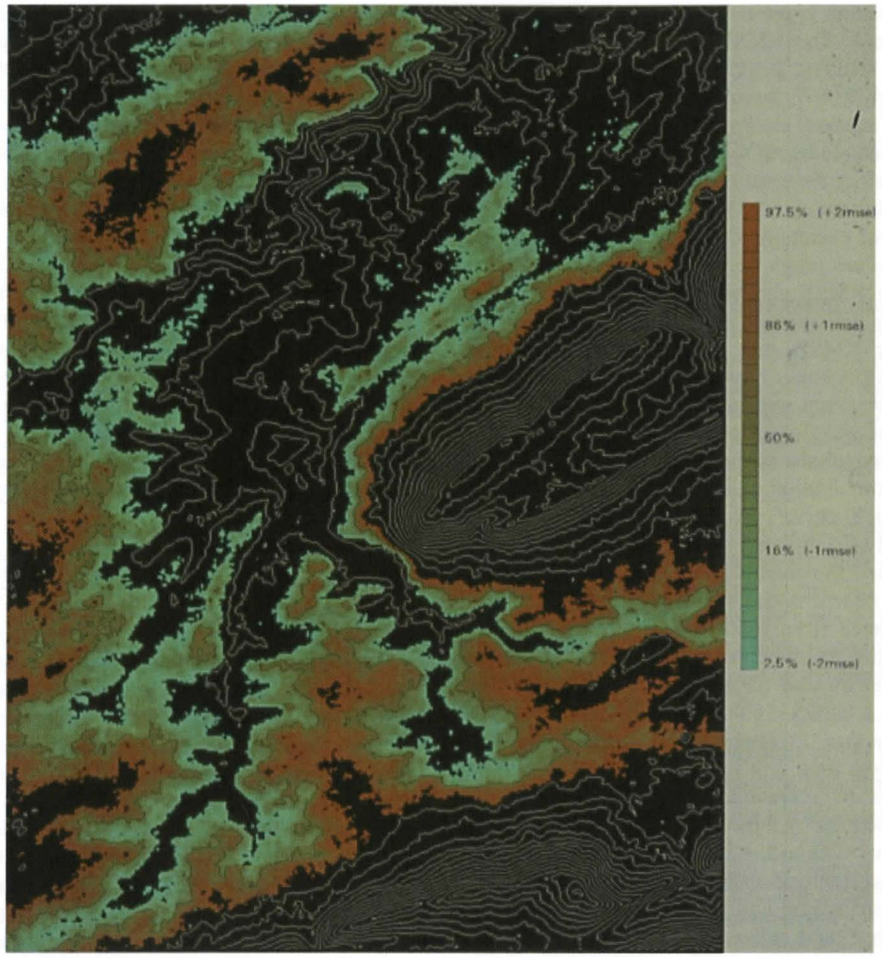


Figure 6.13 Uncertainty in the location of the 350 m contour in the area of State College, Pennsylvania, generated from a US Geological Survey DEM with an assumed RMSE of 7 m. According to the Gaussian distribution with a mean of 350 m and a standard deviation of 7 m, there is a 95% probability that the true location of the 350 m contour lies in the colored area, and a 5% probability that it lies outside (*Source*: Hunter G. J. and Goodchild M. F. 1995 'Dealing with error in spatial databases: a simple case study'. *Photogrammetric Engineering and Remote Sensing* **61**: 529–37)

and it can be generalized to the three-dimensional case. Normally, we would expect the RMSEs of x and y to be the same, but z is often subject to errors of quite different magnitude, for example in the case of determinations of position using GPS. The bivariate Gaussian distribution also allows for correlation between the errors in x and y, but normally there is little reason to expect correlations.

Because it involves two variables, the bivariate Gaussian distribution has somewhat different properties from the simple (univariate) Gaussian distribution. 68% of cases lie within one standard deviation for the univariate case (Figure 6.12). But in the bivariate case with equal standard errors in x and y, only 39% of cases lie within a circle of this radius. Similarly, 95% of cases lie within two standard deviations for the univariate distribution, but it is necessary to go to a circle of radius equal to 2.15 times the x or y standard deviations to enclose 90% of the bivariate distribution, and 2.45 times standard deviations for 95%.

National Map Accuracy Standards often prescribe the positional errors that are allowed in databases. For example, the 1947 US National Map Accuracy Standard specified that 95% of errors should fall below 1/30 inch (0.85 mm) for maps at scales of 1:20 000 and finer (more detailed), and 1/50 inch (0.51 mm) for other maps (coarser, less detailed, levels of granularity than 1:20000). A convenient rule of thumb is that positions measured from maps are subject to errors of up to 0.5 mm at the scale of the map. Table 6.3 shows the distance on the ground corresponding to 0.5 mm for various common map scales.

A useful rule of thumb is that features on maps are positioned to an accuracy of about 0.5 mm.

6.3.4 The spatial structure of errors

The confusion matrix, or more specifically a single row of the matrix, along with the Gaussian distribution, provide convenient ways of describing the error present in a single observation of a nominal or interval/ratio measurement respectively. When a GIS is used to respond to a simple query, such as 'tell me the class of soil at this point', or 'what is the elevation here?', then these methods are good ways of describing the uncertainty inherent in the response. For example, a GIS might respond to the first query with the information 'Class A, with a 30% probability of Class C', and to the second query with the information '350 m, with an RMSE of 7 m'. Notice how this makes it possible to describe nominal data as accurate to a percentage, but it makes no sense to describe a DEM, or any measurement on an interval/ratio scale, as accurate to a percentage. For example, we cannot meaningfully say that a DEM is '90% accurate'.

However, many GIS operations involve more than the properties of single points, and this makes the analysis of error much more complex. For example, consider the query 'how far is it from this point to that point?' Suppose the two points are both subject to error of position, because their positions have been measured using GPS units with mean distance errors of 50 m. If the two measurements were taken some time apart, with different combinations of satellites above the horizon, it is likely that the errors are independent of each other, such that one error might be 50 m in the direction of North, and the other 50 m in the direction of South. Depending on the locations of the two points, the error in distance might be as high as +100 m. On the other hand, if the two measurements were made close together in time, with the same satellites above the horizon, it is likely that the two errors would be similar, perhaps 50 m North and 40 m North, leading to an error of only 10 m in the determination of distance. The difference between these two situations can be measured in terms of the degree of *spatial autocorrelation*, or the interdependence of errors at different points in space (Section 4.6).

The spatial autocorrelation of errors can be as important as their magnitude in many GIS operations.

Table 6.3 A useful rule of thumb is that positions measured from maps are accurate to about 0.5 mm on the map. Multiplying this by the scale of the map gives the corresponding distance on the ground

Map scale	Ground distance corresponding to 0.5 mm map distance
1:1250	0.625 m
1:2500	1.25 m
1:5000	2.5 m
1:10 000	5 m
1:24 000	12 m
1:50 000	25 m
1:100 000	50 m
1:250 000	125 m
1:1 000 000	500 m
1:10 000 000	5 000 m

Spatial autocorrelation is also important in errors in nominal data. Consider a field that is known to contain a single crop, perhaps wheat. When seen from above, it is possible to confuse wheat with other crops, so there may be error in the crop type assigned to points in the field. But since the field has only one crop, we know that such errors are likely to be strongly correlated. Spatial autocorrelation is almost always present in errors to some degree, but very few efforts have been made to measure it systematically, and as a result it is difficult to make good estimates of the uncertainties associated with many GIS operations.

An easy way to visualize spatial autocorrelation and interdependence is through animation. Each frame in the animation is a single possible map, or *realization* of the error process. If a point is subject to uncertainty, each realization will show the point in a different possible location, and a sequence of images will show the point shaking around its mean position. If two points have perfectly correlated positional errors, then they will appear to shake in unison, as if they were at the ends of a stiff rod. If errors are only partially correlated, then the system behaves as if the connecting rod were somewhat elastic.

The spatial structure or autocorrelation of errors is important in many ways. DEM data are often used to estimate the slope of terrain, and this is done by comparing elevations at points a short distance apart. For example, if the elevations at two points 10 m apart are 30 m and 35 m respectively, the slope along the line between them is 5/10, or 0.5. (A somewhat more complex method is used in practice, to estimate slope at a point in the x and y directions in a DEM raster, by analyzing the elevations of nine points – the point itself and its eight neighbors. The equations in Section 14.4 detail the procedure.)

Now consider the effects of errors in these two elevation measurements on the estimate of slope. Suppose the first point (elevation 30 m) is subject to an RMSE of 2 m, and consider possible true elevations of 28 m and 32 m. Similarly the second point might have true elevations of 33 m and 37 m. We now have four possible combinations of values, and the corresponding estimates of slope range from $(33-32)/10 = 0.1$ to $(37-28)/10 = 0.9$. In other words, a relatively small amount of error in elevation can produce wildly varying slope estimates.

The spatial autocorrelation between errors in geographic databases helps to minimize their impacts on many GIS operations.

What saves us in this situation, and makes estimation of slope from DEMs a practical proposition at all, is spatial autocorrelation among the errors. In reality, although DEMs are subject to substantial errors in absolute elevation, neighboring points nevertheless tend to have similar errors, and errors tend to persist over quite large areas. Most of the sources of error in the DEM production process tend to produce this kind of persistence of error over space, including errors due to misregistration of aerial photographs. In other words, errors in DEMs exhibit strong positive spatial autocorrelation.

Another important corollary of positive spatial autocorrelation can also be illustrated using DEMs. Suppose an area of low-lying land is inundated by flooding, and our task is to estimate the area of land affected. We are asked to do this using a DEM, which is known to have an RMSE of 2 m (compare Figure 6.13). Suppose the data points in the DEM are 30 m apart, and preliminary analysis shows that 100 points have elevations below the flood line. We might conclude that the area flooded is the area represented by these 100 points, or 900 × 100 sq m, or 9 hectares. But because of errors, it is possible that some of this area is actually above the flood line (we will ignore the possibility that other areas outside this may also be below the flood line, also because of errors), and it is possible that *all* of the area is above. Suppose the recorded elevation for each of the 100 points is 2 m below the flood line. This is one RMSE (recall that the RMSE is equal to 2 m) below the flood line, and the Gaussian distribution tells us that the chance that the true elevation is actually above the flood line is approximately 16% (see Figure 6.12). But what is the chance that *all* 100 points are actually above the flood line?

Here again the answer depends on the degree of spatial autocorrelation among the errors. If there is none, in other words if the error at each of the 100 points is independent of the errors at its neighbors, then the answer is $(0.16)^{100}$, or 1 chance in 1 followed by roughly 70 zeroes. But if there is strong positive spatial autocorrelation, so strong that all 100 points are subject to exactly the same error, then the answer is 0.16. One way to think about this is in terms of *degrees of freedom*. If the errors are independent, they can vary in 100 independent ways, depending on the error at each point. But if they are strongly spatially autocorrelated, the effective number of degrees of freedom is much less, and may be as few as 1 if all errors behave in unison. Spatial autocorrelation has the effect of reducing the number of degrees of freedom in geographic data below what may be implied by the volume of information, in this case the number of points in the DEM.

Spatial autocorrelation acts to reduce the effective number of degrees of freedom in geographic data.

6.4 U3: Further uncertainty in the analysis of geographic phenomena

6.4.1 Internal and external validation through spatial analysis

In Chapter 1 we identified one remit of GIS as the resolution of scientific or decision-making problems through spatial analysis, which we defined in Section 1.7 as 'the process by which we turn raw spatial data into useful spatial information'. Good science needs secure foundations, yet Sections 6.2 and 6.3 have shown the conception and measurement of many geographic phenomena to be inherently uncertain. How can the outcome of spatial analysis be meaningful if it has such uncertain foundations?

Uncertainties in data lead to uncertainties in the results of analysis.

Once again, there are no easy answers to this question, although we can begin by examining the consequences of accommodating possible errors of positioning, or of aggregating clearly defined units of analysis into artificial geographic individuals (as when people are aggregated by census tracts, or disease incidences are aggregated by county). In so doing, we will illustrate how potential problems might arise, but will not present any definitive solutions – for the simple reason that the truth is inherently uncertain. The conception, measurement, and representation of geographic individuals may distort the outcome of spatial analysis by masking or accentuating apparent variation across space, or by restricting the nature and range of questions that can be asked of the GIS.

There are three ways of dealing with this risk. First, although we can only rarely tackle the *source* of distortion (we are rarely empowered to collect new, completely disaggregate data, for example), we can quantify the way in which it is likely to *operate* (or *propagates*) within the GIS, and can gauge the magnitude of its likely impacts. Second, although we may have to work with aggregated data, GIS allows us to model within-zone spatial distributions in order to ameliorate the worst effects of artificial zonation. Taken together, GIS allows us to gauge the effects of scale and aggregation through simulation of different possible outcomes. This is *internal validation* of the effects of scale, point placement, and spatial partitioning.

Because of the power of GIS to merge diverse data sources, it also provides a means of *external validation* of the effects of zonal averaging. In today's advanced GIService economy (Section 1.5.3), there may be other data sources that can be used to gauge the effects of aggregation upon our analysis. In Chapter 13 we will refine the basic model that was presented in Figure 6.1 to consider how GIS provides a medium for visualizing models of spatial distributions and patterns of homogeneity and heterogeneity.

GIS gives us maximum flexibility when working with aggregate data, and helps us to validate our data with reference to other available sources.

6.4.2 Internal validation: error propagation

The examples of Section 6.3.4 are cases of error propagation, where the objective is to measure the effects of known levels of data uncertainty on the outputs of

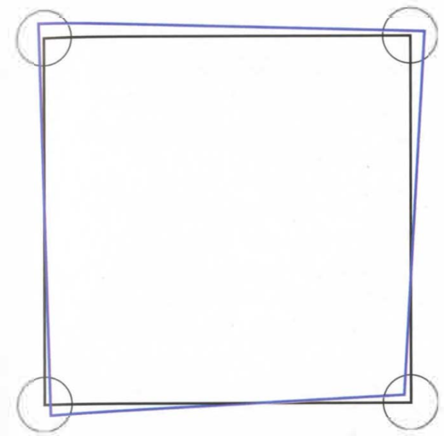


Figure 6.14 Error in the measurement of the area of a square 100 m on each side. Each of the four corner points has been surveyed; the errors are subject to bivariate Gaussian distributions with standard deviations in x and y of 1 m (dashed circles). The blue polygon shows one possible surveyed square (one *realization* of the error model)

GIS operations. We have seen how the spatial structure of errors plays a role, and how the existence of strong positive spatial autocorrelation reduces the effects of uncertainty upon estimates of properties such as slope or area. Yet the cumulative effects of error can also produce impacts that are surprisingly large, and some of the examples in this section have been chosen to illustrate the substantial uncertainties that can be produced by apparently innocuous data errors.

Error propagation measures the impacts of uncertainty in data on the results of GIS operations.

In general two strategies are available for evaluating error propagation. The examples in the previous section were instances in which it was possible to obtain a complete description of error effects based upon known measures of likely error. These enable a complete analysis of uncertainty in slope estimation, and can be applied in the DEM flooding example described in Section 6.3.4. Another example that is amenable to analysis is the calculation of the area of a polygon given knowledge of the positional uncertainties of its vertices.

For example, Figure 6.14 shows a square approximately 100 m on each side. Suppose the square has been surveyed by determining the locations of its four corner points using GPS, and suppose the circumstances of the measurements are such that there is an RMSE of 1 m in both coordinates of all four points, and that errors are independent. Suppose our task is to determine the area of the square. A GIS can do this easily, using a standard algorithm (see Figure 14.9). Computers are precise (in the sense of Section 6.3.2.2 and Box 6.3), and capable of working to many significant digits, so the calculation might be reported by printing out a number to eight digits, such as 10014.603 sq m, or even more. But the number of significant digits will have been determined by the precision of the machine, and not by the accuracy of the determination. Box 6.3 summarized some simple rules for ensuring that the precision used to report a measurement reflects as far as possible its accuracy, and clearly those rules will have been violated if the area is reported to eight digits. But what is the appropriate precision?

In this case we can determine exactly how positional accuracy affects the estimate of area. It turns out that area has an error distribution which is Gaussian, with a standard deviation (RMSE) in this case of 200 sq m – in other words, each attempt to measure the area will give a different result, the variation between them having a standard deviation of 200 sq m. This means that the five rightmost digits in the estimate are spurious, including two digits to the left of the decimal point. So if we were to follow the rules of Box 6.3, we would print 10 000 rather than 10014.603 (note the problem with standard notation here, which does not let us omit digits to the left of the decimal point even if they are spurious, and so leaves some uncertainty about whether the tens and units digits are certain or not – and note also the danger that if the number is printed as an integer it may be interpreted as exactly the whole number). We can also turn the question around and ask how accurately the points would have to be measured to justify eight digits, and the answer is approximately 0.01 mm, far beyond the capabilities of normal surveying practice.

Analysis can be applied to many other kinds of GIS analysis, and Gerard Heuvelink (Box 6.4) discusses several further examples in his excellent text on error propagation in GIS. But analysis is a difficult strategy when spatial autocorrelation of errors is present, and many problems of error propagation in GIS are not amenable to analysis. This has led many researchers to explore a more general strategy of simulation to evaluate the impacts of uncertainty on results.

In essence, simulation requires the generation of a series of realizations, as defined earlier, and it is often called Monte Carlo simulation in reference to the realizations that occur when dice are tossed or cards are dealt in various games of chance. For example, we could simulate error in a single measurement from a DEM by generating a series of numbers with a mean equal to the measured elevation, and a standard deviation equal to the known RMSE, and a Gaussian distribution. Simulation uses everything that is known about a situation, so if any additional information is available we would incorporate it in the simulation. For example, we might know that elevations must be whole numbers of meters, and would simulate this by rounding the numbers obtained from the Gaussian distribution. With a mean of 350 m and an RMSE of 7 m the results of the simulation might be 341, 352, 356, 339, 349, 348, 355, 350, . . .

Simulation is an intuitively simple way of getting the uncertainty message across.

Because of spatial autocorrelation, it is impossible in most circumstances to think of databases as decomposable into component parts, each of which can be independently disturbed to create alternative realizations, as in the previous example. Instead, we have to think of

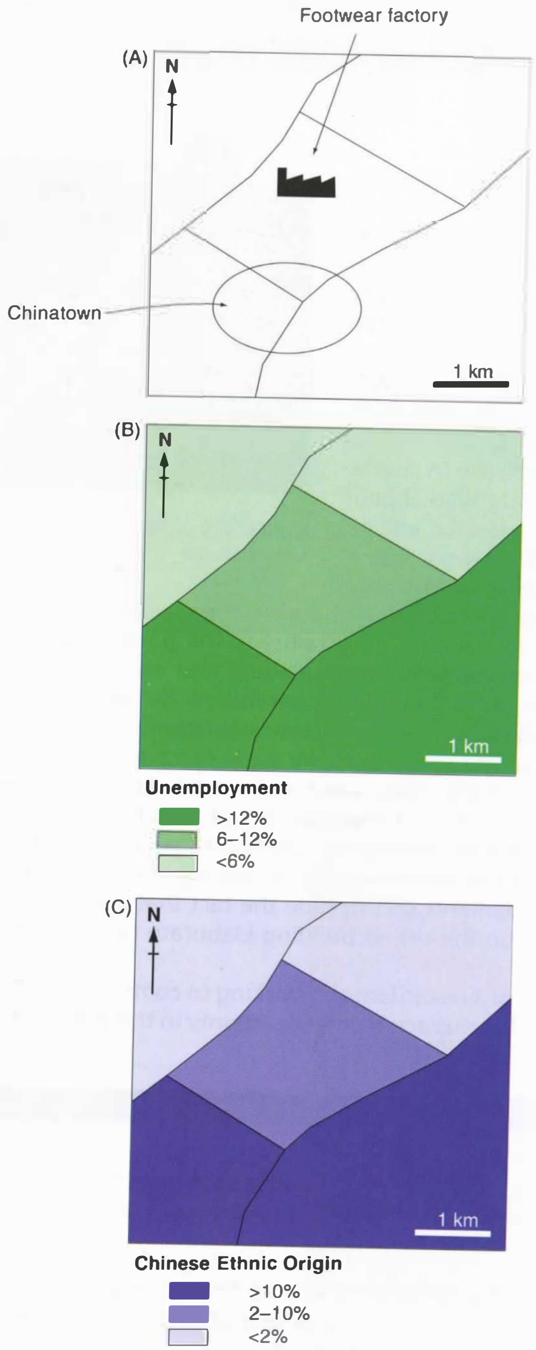


Figure 6.17 The problem of ecological fallacy. Before it closed down, the Anytown footwear factory drew its labor from blue-collar neighborhoods in its south and west sectors. Its closure led to high local unemployment, but not amongst the residents of Chinatown, who remain employed in service industries. Yet comparison of choropleth maps B and C suggests a spurious relationship between Chinese ethnicity and unemployment

Openshaw applied correlation and regression analysis to the attributes of a succession of zoning schemes. He demonstrated that the constellation of elemental zones within aggregated areal units could be used to manipulate the results of spatial analysis to a wide range of quite different prespecified outcomes. These numerical experiments have some sinister counterparts in the real world, the most notorious example of which is the political gerrymander of 1812 (see Section 14.3.2). Chance or design might therefore conspire to create apparent spatial distributions which are unrepresentative of the scale and configuration of real-world geographic phenomena. The outcome of multivariate spatial analysis is also similarly sensitive to the particular zonal scheme that is used. Taken together, the effects of scale and aggregation are generally known as the *Modifiable Areal Unit Problem* (MAUP).

The ecological fallacy and the MAUP have long been recognized as problems in applied spatial analysis and, through the concept of spatial autocorrelation (Section 4.3), they are also understood to be related problems. Increased technical capacity for numerical processing and innovations in scientific visualization have refined the quantification and mapping of these measurement effects, and have also focused interest on the effects of within-area spatial distributions upon analysis.

6.4.4 External validation: data integration and shared lineage

Goodchild and Longley (1999) use the term *concatenation* to describe the integration of two or more different data sources, such that the contents of each are accessible in the product. The polygon overlay operation that will be discussed in Section 14.4.3, and its field-view counterpart, is one simple form of concatenation. The term *conflation* is used to describe the range of functions that attempt to overcome differences between datasets, or to merge their contents (as with rubber-sheeting: see Section 9.3.2.3). Conflation thus attempts to replace two or more versions of the same information with a single version that reflects the pooling, or weighted averaging, of the sources.

The individual items of information in a single geographic dataset often share lineage, in the sense that more than one item is affected by the same error. This happens, for example, when a map or photograph is registered poorly, since all of the data derived from it will have the same error. One indicator of shared lineage is the persistence of error – because all points derived from the same misregistration will be displaced by the same, or a similar, amount. Because neighboring points are more likely to share lineage than distant points, errors tend to exhibit strong positive spatial autocorrelation.

Conflation combines the information from two data sources into a single source.

When two datasets that share no common lineage are concatenated (for example, they have not been subject to the same misregistration), then the relative positions of objects inherit the absolute positional errors of both, even over the shortest distances. While the shapes of objects in each dataset may be accurate, the relative locations of pairs of neighboring objects may be wildly inaccurate when drawn from different datasets. The anecdotal history of GIS is full of examples of datasets which were perfectly adequate for one application, but which failed completely when an application required that they be merged with some new dataset that had no common lineage. For example, merging GPS measurements of point positions with streets derived from the US Bureau of the Census TIGER files may lead to surprises where points appear on the wrong sides of streets. If the absolute positional accuracy of a dataset is 50 m, as it is with parts of the TIGER database, points located less than 50 m from the nearest street will frequently appear to be misregistered.

Datasets with different lineages often reveal unsuspected errors when overlaid.

Figure 6.18 shows an example of the consequences of overlaying data with different lineages. In this case, two datasets of streets produced by different commercial vendors using their own process fail to match in position by amounts of up to 100 m, and also fail to match in the names of many streets, and even the existence of streets.

The integrative functionality of GIS makes it an attractive possibility to generate multivariate indicators from diverse sources. Yet such data are likely to have been collected at a range of different scales, and for a range of areal units as diverse as census tracts, river catchments, land ownership parcels, travel-to-work areas, and market research surveys. Established procedures of statistical inference can only be used to reason from representative samples to the populations from which they were drawn. Yet these procedures do not regulate the assignment of inferred values to (usually smaller) zones, or their apportionment to *ad hoc* regional categorizations. There is an emergent tension within the socio-economic realm, for there is a limit to the uses of inferences drawn from conventional, scientifically valid data sources which are

Figure 6.18 Overlay of two street databases for part of Goleta, California, USA. The red and green lines fail to match by as much as 100 m. Note also that in some cases streets in one dataset fail to appear in the other, or have different connections. The background is dark where the fit is best and white where it is poorest (it measures the average distance locally between matched intersections)

frequently out-of-date, zonally coarse, and irrelevant to what is happening in modern societies. Yet the alternative of using new rich sources of marketing data may be profoundly unscientific in its inferential procedures.

6.4.5 Internal and external validation; induction and deduction

Reformulation of the MAUP into a *geocomputational* (Box 1.9 and Section 16.1) approach to zone design has been one of the key contributions of geographer Stan Openshaw. Central to this is inductive use of GIS to seek patterns through repeated scaling and aggregation experiments, alongside much better external validation, deduced using the multitude of new datasets that are a hallmark of the information age.

The Modifiable Areal Unit Problem can be investigated through simulation of large numbers of alternative zoning schemes.

Neither of these approaches, used in isolation, is likely to resolve the uncertainties inherent in spatial analysis. Zone design experiments are merely playing with the MAUP, and most of the new sources of external validation are unlikely to sustain full scientific scrutiny, particularly if they were assembled through non-rigorous survey designs. The conception and measurement of elemental zones, the geographic individuals, may be *ad hoc*, but they are rarely wholly random either. Can our recognition and understanding of the empirical effects of the MAUP help us to neutralize its effects? Not really. In measuring the distribution of all possible zonally averaged outcomes ('simple random zoning' in analogy to simple random sampling in Section 4.4), there is no tenable analogy with the established procedures of statistical inference and its concepts of precision and error. And even if there were, as we have seen in Section 4.7 there are limits to the application of classic statistical inference to spatial data.

Zoning seems similar to sampling, but its effects are very different.

The way forward seems to be to complement our new-found abilities to customize zoning schemes in GIS with external validation of data and clearer application-centered thinking about the likely degree of within-zone heterogeneity that is concealed in our aggregated data. In this view, MAUP will disappear if GIS analysts understand the particular areal units that they wish to study. There is also a sense here that resolution of the MAUP requires acknowledgment of the uniqueness of places. There is also a practical recognition that the areal objects of study are ever-changing, and our perceptions of what constitutes their appropriate definition will change. And finally, within the socio-economic realm, the act of defining zones can also be self-validating if the allocation of individuals affects the interventions they receive (be they a mail-shot about a shopping opportunity or aid under an areal policy intervention). Spatial discrimination affects

Applications Box 6.5

Uncertainty and town center definition

Although the locus of retail activity in many parts of the US long ago shifted to suburban locations, traditional town centers ('downtowns') remain vibrant and are cherished in most of the rest of the world. Indeed, many nations vigorously defend existing retail centers and through various planning devices seek to regulate 'out of center' development.

Therefore, many interests in the planning and retail sectors are naturally concerned with learning the precise extent of existing town centers. The pressure to devise standard definitions across its national jurisdiction led the UK's central government planning agency (the Office of the Deputy Prime Minister) to initiate a five-year research program to define and monitor changes in the shape, form, and internal geography of town centers across the nation. The work has been based at the Centre for Advanced Spatial Analysis (CASA) at University College London.

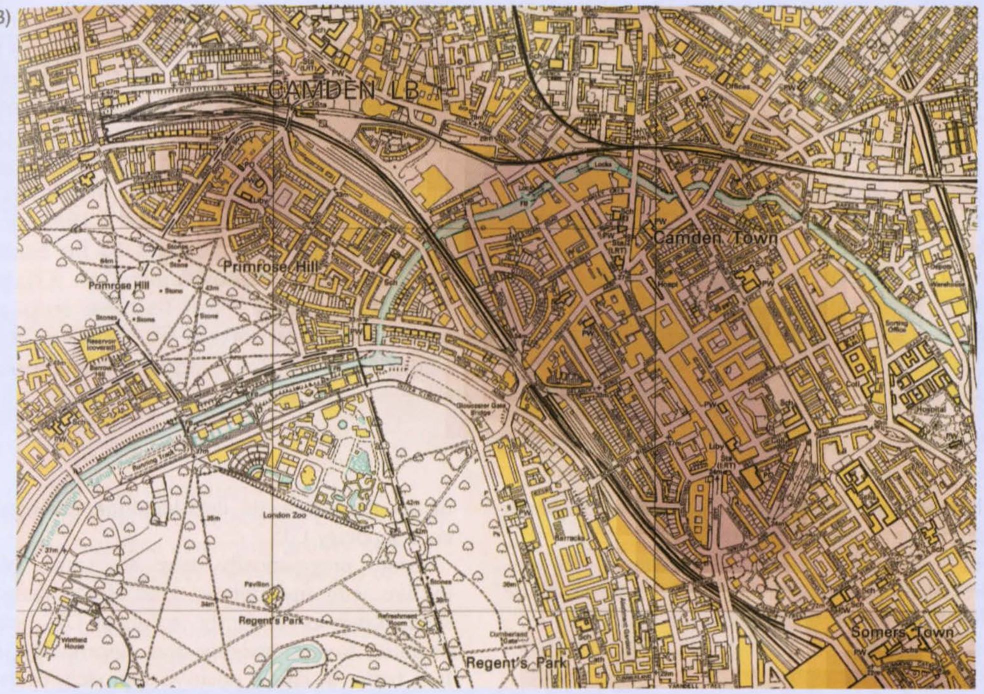
Town centers present classic examples of geographic phenomena with uncertain boundaries. Moreover the extent of any given town center is likely to change over time – for example, in response to economic fortunes consequent upon national business cycles. Candidate *indicator* variables of town centeredness might include tall buildings, pedestrian traffic, high levels of retail employment, and high retail floorspace figures.

After a consultation period with user groups (in the spirit of public participation in GIS, PPGIS: see Section 13.4) a set of the most pertinent indicators was agreed (the *conception* stage). These indicator variables measured retail and hospitality industry employment, shop and office floorspace, and retail, leisure, and service employment. The indicator measures were standardized, weighted, and summed into a summary index *measure*. This measure, mapped for all town centers, was the principal deliverable

(A)

Figure 6.19 (A) Camden Town Center, London (Reproduced by permission of Jamie G.A. Quinn); (B) a data surface representing the index of town center activity (the darker shades of red indicate greater levels of retail activity); and (C) the Camden Town Center report: Camden Town center boundary is blue, whilst the orange lines denote the retail core of Camden and of nearby town centers – the darker shades of red again indicate greater levels of retail activity

(B)



(C)

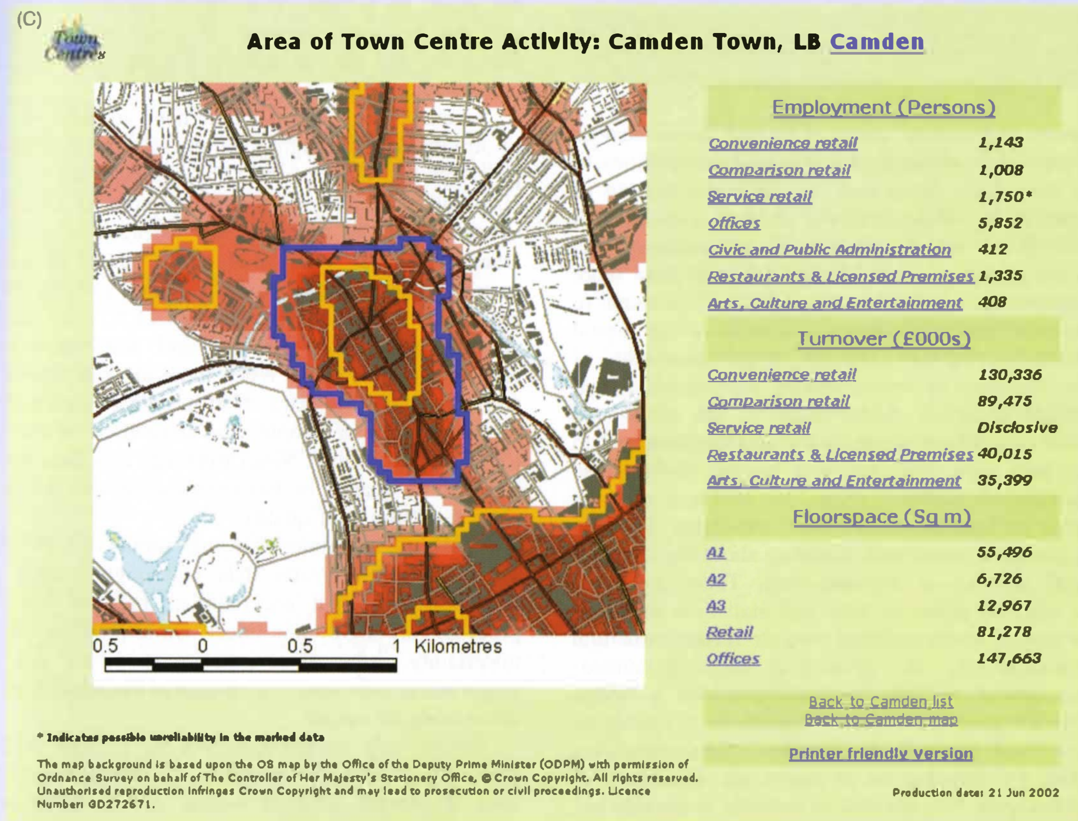


Figure 6.19 *(continued)*

of the research. After further consultation, the CASA team chose to represent the 'degree of town centeredness' as a field variable. This choice reflected various priorities, including the need to maintain confidentiality of those data that were not in the public domain, an attempt to avoid the worst effects of the MAUP, and the need to communicate the rather complex concept of the 'degree of town centeredness' to an audience of 'spatially *un*aware professionals' (see Section 1.4.3.2). The datasets used in the projects each represent populations (not samples), and so kernel density estimation (Section 14.4.4.4) was used to create the composite surfaces: the size of the kernel was subjectively set at 300 meters, because of the resolution of the data and on the basis of the empirical observation that this is the maximum distance that most shoppers are prepared to walk.

An example of the composite surface 'index of town center activity' for the town center of Camden, London (Figure 6.19A) is shown in Figure 6.19B. For reasons to be explored in our discussion of geovisualization (Chapter 13), most users prefer maps to have crisp and not graduated or uncertain boundaries. Thus crisp boundaries were subsequently interpolated as shown in Figure 6.19C.

spatial behavior, and so the principles of zone design are of much more than academic interest.

Many of issues of uncertainty in conception, measurement, representation, and analysis come together in the definition of town center boundaries (see Box 6.5).

6.5 Consolidation

Uncertainty is certainly much more than error. Just as the amount of available digital data and our abilities to process them have developed, so our understanding of the quality of digital depictions of reality has broadened. It is one of the supreme ironies of contemporary GIS that as we accrue more and better data and have more computational power at our disposal, so we seem to become more uncertain about the quality of our digital representations and the adequacy of our areal units of analysis. Richness of representation and computational power only make us more aware of the range and variety of established uncertainties, and challenge us to integrate new ones. The only way beyond this impasse is to advance hypotheses about the structure of data, in a spirit of humility rather than conviction. But this implies greater *a priori* understanding about the structure in spatial as well as attribute data. There are some general rules to guide us here and statistical measures such as spatial autocorrelation provide further structural clues (Section 4.3). The developing range of context-sensitive spatial analysis methods provides a bridge between such general statistics and methods of specifying place or local (natural) environment. Geocomputation helps too, by allowing us to assess the sensitivity of outputs to inputs, but, unaided, is unlikely to provide any unequivocal best solution. The fathoming of uncertainty requires a combination of the cumulative development of *a priori* knowledge (we should expect scientific research to be cumulative in its findings), external validation of data sources, and inductive generalization in the fluid, eclectic data-handling environment that is contemporary GIS.

More pragmatically, here are some rules for how to live with uncertainty: First, since there can be no such thing as perfectly accurate GIS analysis, it is essential to acknowledge that uncertainty is inevitable. It is better to take a positive approach, by learning what one can about uncertainty, than to pretend that it does not exist. To behave otherwise is unconscionable, and can also be very expensive in terms of lawsuits, bad decisions, and the unintended consequences of actions (see Chapter 19).

Second, GIS analysts often have to rely on others to provide data, through government-sponsored mapping programs like those of the US Geological Survey or the UK Ordnance Survey, or commercial sources. Data should never be taken as the truth, but instead it is essential to assemble all that is known about the quality of data, and to use this knowledge to assess whether, actually, the data are fit for use. Metadata (Section 11.2.1) are designed specifically for this purpose, and will often include assessments of quality. When these are not present, it is worth spending the extra effort to contact the creators of the data, or other people who have tried to use them, for advice on quality. Never trust data that have not been assessed for quality, or data from sources that do not have good reputations for quality.

Third, the uncertainties in the outputs of GIS analysis are often much greater than one might expect given knowledge of input uncertainties, because many GIS processes are highly non-linear. Other processes dampen uncertainty, rather than enhance it. Given this, it is important to gain some impression of the impacts of input uncertainty on output.

Fourth, rely on multiple sources of data whenever you can. It may be possible to obtain maps of an area at several different scales, or to obtain several different vendors' databases. Raster and vector datasets are often complementary (e.g., combine a remotely sensed image with a topographic map). Digital elevation models can often be augmented with spot elevations, or GPS measurements.

Finally, be honest and informative in reporting the results of GIS analysis. It is safe to assume that GIS designers will have done little to help in this respect – results will have been reported to high apparent precision, with more significant digits than are justified by actual accuracy, and lines will have been drawn on maps with widths that reflect relative importance, rather than uncertainty of position. It is up to you as the user to redress this imbalance, by finding ways of communicating what you know about accuracy, rather than relying on the GIS to do so. It is wise to put plenty of caveats into reported results, so that they reflect what you believe to be true, rather than what the GIS appears to be saying. As someone once said, when it comes to influencing people 'numbers beat no numbers every time, whether or not they are right', and the same is certainly true of maps (see Chapters 12 and 13).

Questions for further study

1. What tools do GIS designers build into their products to help users deal with uncertainty? Take a look at your favorite GIS from this perspective. Does it allow you to associate metadata about data quality with datasets? Is there any support for propagation of uncertainty? How does it determine the number of significant digits when it prints numbers? What are the pros and cons of including such tools?
2. Using aggregate data for Iowa counties, Openshaw found a strong positive correlation between the proportion of people over 65 and the proportion who were registered voters for the Republican party. What if anything does this tell us about the tendency for older people to register as Republicans?
3. Find out about the five components of data quality used in GIS standards, from the information available at **www.fdgc.gov**. How are the five components applied in the case of a standard mapping agency data product, such as the US Geological Survey's Digital Orthophoto Quarter–Quadrangle program? (Search the Web for the appropriate documents.)
4. You are a senior retail analyst for Safemart, which is contemplating expansion from its home US state to three others in the Union. Assess the relative merits of your own company's store loyalty card data (which you can assume are similar to those collected by any retail chain with which you are familiar) and of data from the 2000 Census in planning this strategic initiative. Pay, particular attention to issues of survey content, the representativeness of population characteristics, and problems of scale and aggregation. Suggest ways in which the two data sources might complement one another in an integrated analysis.

Further reading

Burrough P.A. and Frank A.U. (eds) 1996 *Geographic Objects with Indeterminate Boundaries*. London: Taylor and Francis.

Fisher P.F. 2005 'Models of uncertainty in spatial data.' In Longley P.A., Goodchild M.F., Maguire D.J. and Rhind D.W. (eds) *Geographical Information Systems: Principles, Techniques, Management and Applications* (abridged edition). Hoboken, N.J.: Wiley, pp. 191–205.

Goodchild M.F. and Longley P.A. 2005 'The future of GIS and spatial analysis.' In Longley P.A., Goodchild M.F., Maguire D.J. and Rhind D.W. (eds) *Geographical Information Systems: Principles, Techniques, Management and Applications* (abridged edition). Hoboken, N.J.: Wiley, pp. 567–80.

Heuvelink G.B.M. 1998 *Error Propagation in Environmental Modelling with GIS*. London: Taylor and Francis.

Openshaw S. and Alvanides S. 2005 'Applying geocomputation to the analysis of spatial distributions'. In Longley P.A., Goodchild M.F., Maguire D.J. and Rhind D.W. (eds) *Geographical Information Systems: Principles, Techniques, Management and Applications* (abridged edition). Hoboken, N.J.: Wiley, pp. 267–282.

Zhang J.X. and Goodchild M.F. 2002 *Uncertainty in Geographical Information*. New York: Taylor and Francis.